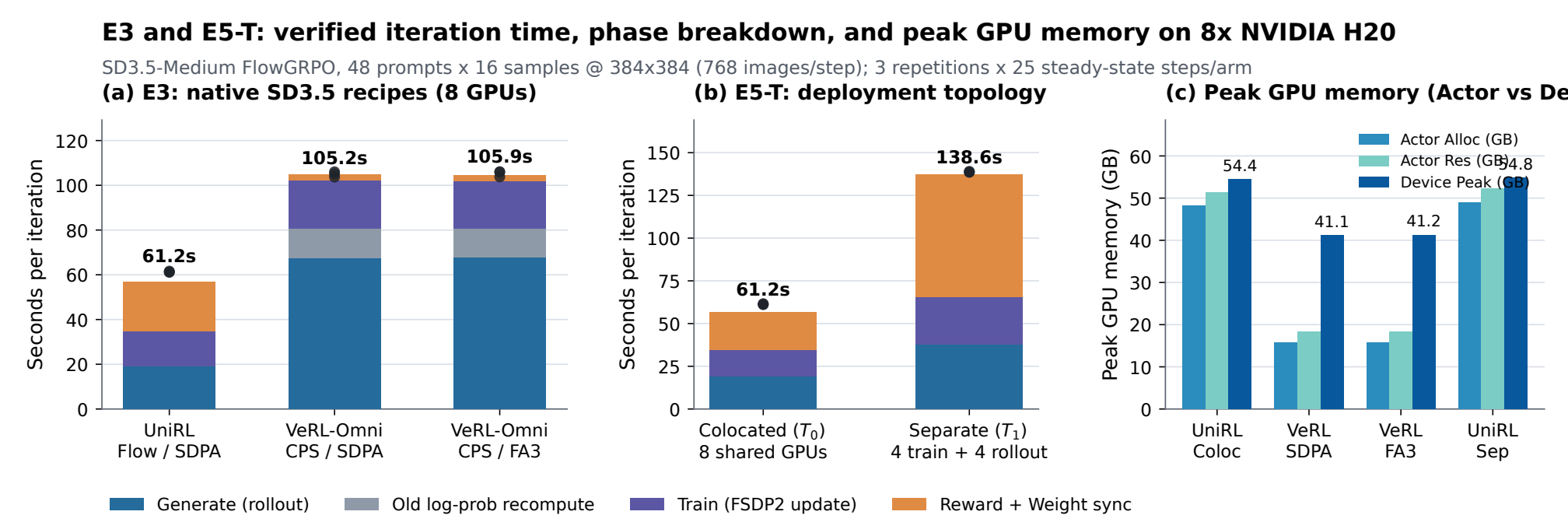
E3 and E5-T: verified iteration time, phase breakdown, and peak GPU memory on 8x NVIDIA H20
SD3.5-Medium FlowGRPO, 48 prompts x 16 samples @ 384x384 (768 images/step); 3 repetitions x 25 steady-state steps/arm
(a) E3: native SD3.5 recipes (8 GPUs)
Seconds per iteration
61.2s
105.2s
105.9s
UniRL
Flow / SDPA
VeRL-Omni
CPS / SDPA
VeRL-Omni
CPS / FA3
(b) E5-T: deployment topology
Seconds per iteration
61.2s
138.6s
Colocated (T_0)
8 shared GPUs
Separate (T_1)
4 train + 4 rollout
(c) Peak GPU memory (Actor vs De
Peak GPU memory (GB)
Actor Alloc (GB)
Actor Res (GB)
Device Peak (GB)
54.4
41.1
41.2
54.8
UniRL
Coloc
VeRL
SDPA
VeRL
FA3
UniRL
Sep
Generate (rollout)
Old log-prob recompute
Train (FSDP2 update)
Reward + Weight sync